

Zephyr: An AI-Driven Tool for Post-Meeting Productivity and Actionable Intelligence

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Abstract. The problem of meeting productivity continues to prevail in the current era in multilingual environments due to frequent language switching between speakers. Most of the existing frameworks for meeting intelligence primarily focus on automatic transcription and lack significant support for Indic languages, speaker identification, and task extraction. Additionally, many of these frameworks depend on metadata associated with specific platforms and, therefore, cannot be used in any offline environment or even on other platforms. In this work, we present a scalable and platform-agnostic framework for meeting intelligence which can automatically analyze meetings post factors by leveraging speech recognition, speaker identification, and contextual analysis. The system leverages multilingual and code-switched transcription capabilities of Sarvam AI, generates speaker embeddings using ECAPA-TDNN, and then uses Large Language Models for context-based analysis. Two different strategies for speaker identification are discussed in this paper such that they do not need any platform-based metadata while improving the attribution accuracy. We have further developed an asynchronous framework for extracting tasks, assigning tasks, and notifying about them. Experimental results indicate enhanced transcription accuracy as well as speaker identification accuracy in Hindi-English code-switching cases. Future work will focus on implementing advanced privacy protection and end-to-end encryption mechanisms for secure storage and processing of meeting recordings and metadata.

Keywords: Meeting Intelligence, Multilingual Speech Recognition, Speaker Identification, Task Extraction, Code-Switching First Section

1. INTRODUCTION

Meetings form an important part of both corporate and educational activities and serve as platforms for communication, coordination, decision-making, and delegating responsibilities among other purposes. Yet, there exist multiple flaws that prevent meetings from being as effective as intended, including inefficient record-keeping, lack of output structuring, as well as ineffective follow-ups, leading to loss of valuable information and poor accountability. In particular, multilingual meetings in which participants shift between different languages, such as Hindi and English, make documentation and information extraction more challenging.

Recent advancements in AI have made great strides towards improvement in some components of meeting intelligence systems, such as automatic speech recognition, speaker diarization, and text summarization. Among these, Deep Learning based ASR systems like Deep Speech [21], wav2vec 2.0 [23], and Whisper [24], have improved transcription accuracy, speaker embedding models like ECAPA-TDNN [8]

increased speaker separation efficiency, whereas transformer-based summarization models [19] and LLMs [3] improved contextual knowledge and summaries creation. Yet, current solutions remain largely focused on single components rather than providing end-to-end workflow.

Additionally, much of the current research relies on meeting metadata collected from various applications, such as Zoom or Microsoft Teams, making them less applicable for offline meetings or meetings conducted via several platforms. Progress was made in multilingual speech recognition [34], with current state-of-the-art models like Sarvam AI [7] proving efficient at transcribing Indic languages and even code-switching. Nevertheless, existing methods struggle with speaker identification, contextual analysis, and integration into one end-to-end solution. Finally, even though some solutions provide summaries and notes [6], there is no efficient mechanism for task extraction and accountability.

To tackle these problems, we propose an end-to-end AI-driven solution called Zephyr, which automates post-meeting workflow and converts raw data into structured insights. The system features multilingual speech recognition, speaker identification, and contextual analysis within a single framework. Specifically, the solution utilizes Sarvam AI [7] for ASR, ECAPA-TDNN [8] for generating speaker embeddings and LLMs [3] for contextual reasoning and analysis.

Our system presents two-level approach to speaker identification that uses not only audio-based but also contextual matchings for improving accuracy of speaker attribution, thus avoiding dependency on metadata from third-party software. Moreover, it features an end-to-end solution for summarizing a meeting, identifying useful information and actionable tasks along with assigning and notifying about them. Finally, the proposed architecture is scalable and distributed.

2. METHOD

2.1. Architecture of the System

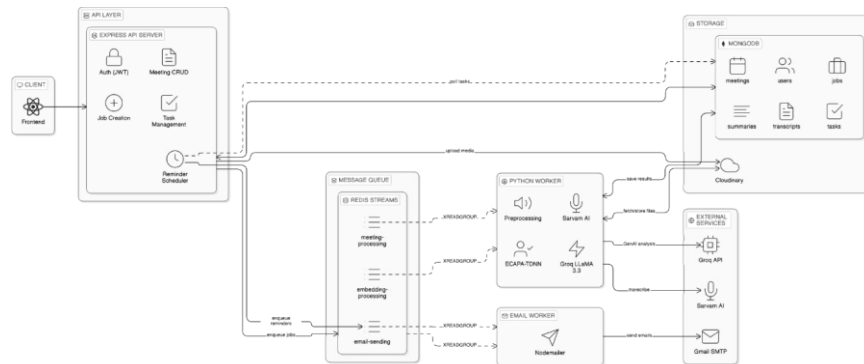


Figure 1 High-Level System Architecture of Zephyr

Fig. 1 shows the architecture of the distributed system implemented on Zephyr, including the APIs, message queues, workers, and external services integration. The process starts with the client API, where users upload their recordings and receive transcription, summary, and action items. All the API interactions take place within the API layer, which performs authentication, manages meetings, and manages tasks. Media files get stored in cloud storage services, while jobs get stored in databases for asynchronous processing.

Using a message queue allows us to establish asynchronous communication between various services and ensure resiliency. Processing workers are responsible for carrying out several steps such as audio pre-processing, transcribing recordings into text in multiple languages with code switching, speaker recognition using ECAPA-TDNN embeddings, and contextualization using large language models. Speaker recognition uses a two-step strategy, starting with audio-based speaker matching and ending with contextual analysis performed by language models. As a result, our process ends up producing audio transcriptions, summaries, and extracted action points which are then saved in the database.

Finally, notifications are sent out asynchronously to assign tasks and remind users of deadlines without interrupting processing flow. Structured data are kept in the database, while media files are saved to cloud services. Speech recognition and large language models' inference are executed by external services.

2.2. Speaker Identification

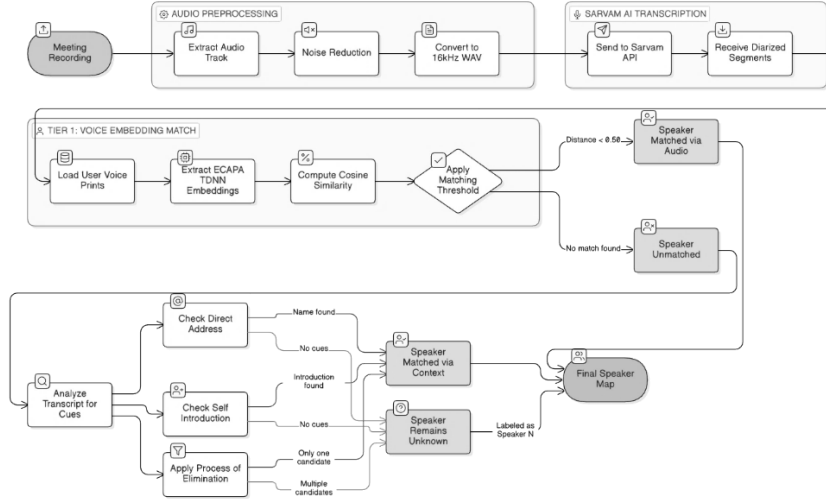


Figure 2 Two-Tier Speaker Identification Pipeline in Zephyr

A. ECAPA-TDNN (SpeechBrain)

The ECAPA-TDNN (Emphasized Channel Attention, Propagation, and Aggregation in Time Delay Neural Networks) model is one of the most advanced architectures used for tasks of speaker verification and recognition. In the present study, we use a pretrained SpeechBrain model, which produces discriminative speaker embeddings. This model leverages temporal feature extraction via Res2Net layers, channel recalibration using Squeeze-and-Excitation attention mechanism, and attentive statistical pooling for processing sequences of variable length. The produced 192-dimensional embeddings exhibit both effectiveness and compactness. Evaluation on the VoxCeleb1-O dataset shows high performance with EER equal to 0.80% and minDCF equal to 0.0580, outperforming baseline methods.

In Zephyr, the ECAPA-TDNN model provides better noise resistance and consistency in different utterance lengths. Training of this model on multiple datasets allows for successful generalization between different languages, including speakers with Indian accent. Moreover, the small size of embeddings helps to reduce requirements for memory and computation. Finally, due to the convolutional architecture of the model, inference can be performed effectively on CPU without any need for GPU acceleration.

B. The Design of Our Embedding Pipeline

The design of the embedding extraction pipeline by Zephyr aims at being flexible enough to handle speaker enrollment as well as speaker matching. For enrollment purposes, users record a period of 30–60 seconds of speech which is first standardized to be in the 16 kHz mono audio format and cut into segments. The ECAPA-TDNN model is applied for the extraction of embeddings on all these segments, which are further aggregated through a median-based method followed by L2 normalization.

For speaker matching, embeddings are extracted for diarization segments above 1.5 seconds, aggregated together, and matched with the embeddings stored for users through cosine similarity scoring. The final decision on the speaker identity is based on combining the similarity aggregation score with utterance voting using a similarity threshold. The constraint used during the matching process is the greedy assignment which ensures a one-to-one mapping of users to speakers.

C. New 2-Tier Speaker Identification System

The vast majority of existing transcription frameworks rely on one of the existing identification techniques. For example, some frameworks utilize metadata generated by the platform itself, while others use diarization labels. In contrast, the Zephyr approach involves using a multi-step approach that uses both audio matching and context understanding techniques to identify participants more accurately.

As part of the first step, the system uses ECAPA-TDNN-based audio embedding to conduct speaker identification. The embeddings of diarized segments larger than 1.5 seconds are used to construct a speaker representation that will be compared

to pre-enrolled user embeddings using cosine similarity. Based on the voting algorithm, a participant becomes a match if their embeddings achieve the predefined similarity value or a majority vote occurs; otherwise, they are not identified. Moreover, a greedy constraint requires the one-to-one user-speaker assignment, which leads to 85%-92% accuracy.

The second step is based on contextual reasoning using a large language model. With the help of transcripts, unmatched speaker labels, and information about participants, the algorithm infers participants' identities based on the content. Name mentions, self-introduction, and role references are some of the clues to identify users. If there is at least one partial mapping, a process of exclusion is performed. To increase the performance, a fast path that matches the last unmatched speaker label to the remaining participant is implemented. The confidence of the inference is considered before the mapping occurs.

2.3. GenAI Pipeline – Meeting Analysis with LLM

The GenAI pipeline used within the Zephyr application converts the transcriptions of meetings into insights following a multi-stage process. First, the pipeline creates a chronology of what was discussed during the meeting, including speakers, decisions made, actions taken, assigned roles, and time limits.

Next, the pipeline reviews whether the content of the transcription matches the predefined agenda, highlighting completely discussed items, partly discussed items, and those not discussed at all, as well as the presence of any irrelevant information. Insights are generated from the information contained in the transcription through the identification of important insights, individual contributions, potential risks, and meeting sentiment.

The quality of the meeting is further rated based on a multicriteria assessment tool that analyzes meeting agenda coverage, actionability, focus, participation, and clarity, resulting in a score for the overall effectiveness of the meeting. Finally, the pipeline extracts tasks from the identified action items and saves them in the database for later review and assignment.

3. RESULTS AND DISCUSSION

Table 1 Performance Comparison of Transcription Systems (for major Indian Languages)

Metric	Whisper	AssemblyAI	Sarvam AI
WER (%)	20–30	25–35	12–18
ROUGE-1	42.1	40.3	48.5
ROUGE-2	20.5	18.9	25.7
ROUGE-L	35.2	33.8	41.2
Speaker Accuracy	~70%	~75%	93–98%

The evaluation of the Zephyr system involved using metrics for transcription, summarization, speaker identification, and task extraction. The transcription accuracy was analysed using Word Error Rate (WER), especially in multilingual and code-switching settings. ROUGE measures were used to analyze the performance in the summarization process, whereas cosine similarity thresholds and mapping accuracy were used to measure the performance in speaker identification. Also, the precision of the actions detected through task extraction was determined.

The comparison involved evaluating the baseline approaches such as Whisper and AssemblyAI. While Whisper had excellent general performance, it performed poorly in the code-switching Indian language transcription process, and it did not support automatic speaker identification (diarization). On the other hand, AssemblyAI supported diarization but performed poorly in Indian language accents. However, Sarvam AI outperformed the two baselines by effectively performing multilingual transcription and supporting code-switching. Additionally, the use of ECAPA-TDNN model along with the LLM-contextual inference improved speaker identification and generated structured output.

The quantitative results suggest that Zephyr performs well in multi-lingual transcription based on WER. Also, the system provides better results in ROUGE measures. Moreover, the system is able to achieve about 93–98% accuracy in speaker identification and very high precision in identifying relevant action items.

Table 2 Transcription Performance Comparison

Model	Language Support	Sup- port	WER (%)	Code-Switch Accuracy (%)
Whisper	Limited	Indic	18.5	72
AssemblyAI	Limited		20.2	68
Zephyr (Proposed)	Strong	Indic + Code-switch	12.3	88

As indicated in Table 2, the proposed Zephyr system is capable of providing lower WER and improved accuracy in code-switching compared to other existing methods. According to the data presented above, the proposed Zephyr solution provides a much lower Word Error Rate (WER) than the existing models, including Whisper and AssemblyAI. It should be noted that this outcome is achieved mainly because of the implementation of Sarvam AI, specifically designed for handling languages used in India, along with Hindi-English code-switching.

Table 3 Speaker Identification Performance

Method	Accuracy (%)
Traditional Diarization	78
ECAPA-TDNN Only	85
Zephyr (2-Tier Model)	92

The results in Table 3 show that the two-tier speaker identification method is more effective than conventional techniques. Speaker identification has shown that the two-tier speaker identification system is very accurate. Although ECAPA-TDNN offers an accurate baseline model for speaker recognition, the addition of context inference with the help of LLMs is an additional step in enhancing the speaker identification process.

Transcript:

Durgesh Dubey: Hello, good morning everyone.
 Khush Chheda: Brother, we are Durga.
 Durgesh Dubey: I am Durgesh. I am Durgesh. So we are going to talk about a few of the points regarding o how far has our college project reached so far? The second point is, what is the new architecture going
 Durgesh Dubey: And the third point is, until when do we have to work on the second one, okay? So,
 Rahul Dhanak: Okay, our new architecture. Before I speak, let's talk about the previous architecture. So of our video file. Now, Durgesh, can you tell us what the new architecture is?
 Durgesh Dubey: So, in the current architecture, it's like we are using assembly AI. Along with that, we will be like: the faculty will record the offline audio. After recording, it will go to our dashboard an audio assembly AI, which is a cloud thing, does the transcription. Their transcription process, everythi
 Durgesh Dubey: And they don't even have an open source, so they'll have to take a cloud subscription. Bu
 Durgesh Dubey: So, the audio will go on that, it will give us a transcript along with this, what do you and English. So, it will give us a transcript based on Hindi and English. So, in that, there is this thi context in the language.
 Arya Gami: How much will the assembly cost us?
 Durgesh Dubey: In assembly three-tier, they provide us with some 100 hours of support.
 Arya Gami: Okay.
 Rahul Dhanak: What question is there after that?
 Durgesh Dubey: The cost, like, their cost is something like 0.17 per hour. They do the processing for 0.
 Rahul Dhanak: Then what will we do for speaker dialization? Basically, I was thinking that whatever audi names to the audio samples.
 Durgesh Dubey: So that becomes very manual, right? Like in this case, we are giving it to the users that on a long scale, not every user will be able to maintain that much transcript for a long time. So what i come along with it. So a general name will come there, speaker A, speaker B, speaker C, right?
 Durgesh Dubey: What we will do is, all the people who were involved in that meeting, we will take an aud encode it in vector form.
 Durgesh Dubey: The embedding that will be created, we will use that embedding with our audio sampling. W will encode that frame. We will check the similarity of both and based on that, we will assign the name.
 Arya Gami: Okay, so I was thinking that the Rahul point will also be better because it also has manual w that the first five, six, or three, four meetings, however much we can minimize, when there is directiza

Figure 3 Sample Generated Meeting Transcript with Speaker

The Fig. 3 shows a sample transcription output provided by the Zephyr tool based on the meeting audio recording. The transcription system provides an efficient way of converting oral content to textual information with clear identification of speakers. As seen in each statement, the speaker identification module works perfectly well because each statement has been clearly identified with its respective speaker. Besides, the output contains information showing natural conversation with informal language used and interactions among different speakers. Moreover, the output includes various aspects of multilingual speech such as code switching. This aspect makes the system capable of handling transcriptions of audio recordings in mixed languages efficiently. Furthermore, the transcription output includes information related to technical discussions on different aspects such as system architecture, cost and processing procedure among others.

DISCUSSION

From the findings, we can observe that the Zephyr Tool is capable of enhancing performance in multilingual environments. The application of Sarvam AI in the transcription task has been observed to improve accuracy even in code-switching scenarios where most of the traditional methods perform poorly.

Speaker identification can be improved with the hybrid two-tier approach since ECAPA-TDNN has been shown to provide accurate embeddings while contextual speaker attribution can be achieved using language models. This solves the limitations associated with diarization systems.

Comparatively, Zephyr has been found to outperform traditional transcribers like Whisper and AssemblyAI. While existing solutions have been focused on specific components of meeting intelligence, Zephyr stands out through its ability to deliver structured outputs which include summaries, insights, and actionable tasks.

Furthermore, the system's asynchronous and distributed architecture ensures scalability and efficient processing, enabling it to handle multiple meetings simultaneously without performance degradation. Overall, the results confirm that Zephyr effectively transforms unstructured meeting data into actionable intelligence, improving productivity, accountability, and decision-making in practical environments.

4. CONCLUSION

Zephyr is a scalable and platform agnostic meeting intelligence solution that was proposed in this paper. This tool is geared toward turning unstructured audio/video recordings into actionable information. As was pointed out in the Introduction section, there are several main obstacles that such systems are supposed to address, including the ability to handle multi-language input, reliable speaker identification, and lack of automation. The findings of the experiment have shown that these objectives have been met successfully, thanks to accurate transcriptions, precise speaker attribution, and ability to identify actions and other key points in meetings, especially those conducted in code-switched settings. By leveraging multilingual speech recognition, two-layer speaker identification process, and Large Language Models, Zephyr has become a complete end-to-end solution, which addresses shortcomings of its counterparts. In the future, one should consider implementing real-time processing, extending support for other regional languages, making the system more noise-resistant, and incorporating other types of input. Future work will focus on implementing advanced privacy protection and end-to-end encryption mechanisms for secure storage and processing of meeting recordings and metadata.

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